

Sohei Yamada

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Education

New York University Brooklyn, NY, USA
Ph.D. in Urban Systems, GPA 4.00/4.00 *Ongoing*

- Provisional thesis title: Urban heat risk mitigation across scales: urban forest ecophysiology, morphology, and climatology. Advisor: Masoud Ghandehari.

University of California, San Diego La Jolla, CA, USA
Master of Public Policy *June 2023*

- Specializing in environmental policy, program design & evaluation, and GIS.

Waseda University Tokyo, Japan
Master of Engineering *March 2015*

- Thesis title: Development of Droplet Transport Model for Critical Power Prediction. Advisor: Shinichi Morooka.

Bachelor of Engineering *March 2013*

- Thesis title: Prediction of a boiling transition location and heated surface temperature for upward flow in a vertical tube. Advisor: Shinichi Morooka.

Research Experience

NYU City Lab Brooklyn, NY, USA
Advisor: Masoud Ghandehari

My research focuses on urban heat risk analytics and mitigation strategies across multiple spatial and temporal scales, integrating computational fluid dynamics, machine learning, and remote sensing with in-situ measurements.

- Developed a weather-aware land-surface-temperature prediction model at 30-meter spatial and 10-minute temporal resolution ($R^2 > 0.9$), combining urban morphology and surface cover features from Landsat and Sentinel-2 with diurnal variation from GOES-16 data, validated against ECOSTRESS observations. Applied the framework to quantify heat-risk inequality across NYC neighborhoods under present-day and CMIP6 climate-change scenarios.
- Developed a transient multiphysics CFD framework incorporating three evapotranspiration models of increasing physical complexity, validated against in-situ microclimatic measurements and infrared surface temperature observations. Simulations of a 0.5-ha urban forest in New York City demonstrate that the stomatal-resistance-based model outperforms empirical formulations, and that evapotranspiration cooling extends approximately 300 m downwind and exceeds 1 K at pedestrian height. Partial correlation analysis identifies relative humidity and wind speed as the dominant meteorological drivers of evapotranspiration cooling power.

Waseda University Thermal Hydraulics Lab

Tokyo, Japan

Advisor: Shinichi Morooka

This research focuses on thermal-hydraulic analysis of nuclear fuel assemblies, with particular emphasis on predicting boiling transition and fuel rod surface temperature under two-phase flow conditions in vertical circular tubes.

- Developed and validated a predictive framework for boiling transition and fuel rod surface temperature in upward two-phase flow within vertical circular tubes. The model improves upon existing correlations for the critical heat flux condition by accounting for local thermal-hydraulic parameters along the heated surface. Simulations were benchmarked against experimental data, demonstrating improved accuracy in predicting the axial location of boiling transition and the resulting surface temperature excursion, with direct implications for safety margin assessment in nuclear reactor fuel assemblies.

**Research
Interest**

My broad research interests are:

- Optimization of climate change adaptation strategies
- Visualization of weather-related risks
- High spatiotemporal resolution acquisition of meteorological data
- Improving accuracy and reducing computational cost in climate risk analysis
- Integration of multi-scale models in climate simulation

Awards

- **Urban Doctoral Fellow.** New York University. 2025.
- **Fellowship Award.** Atomic Energy Society of Japan. 2015.
- **Best Presentation Award.** Atomic Energy Society of Japan, Thermal Hydraulics Division. 2015.
- **Encouragement Award.** Atomic Energy Society of Japan. 2014.
- **Best Presentation Award.** The Japan Society of Mechanical Engineers. 2013.
- **Encouragement Award.** Atomic Energy Society of Japan. 2013.

**Teaching
Experience**

Climate Risk Analysis and Urban Sustainability

Course Assistant

Spring 2025 and 2026

- Teaching assistant for the graduate Climate Risk Analysis and Urban Sustainability class. Topics included fundamentals of climate risk, hurricane and flood modeling, social vulnerability assessment, infrastructure resilience, climate adaptation strategies, and decision-making under uncertainty.
- Developed and delivered advanced instruction in GIS and Python programming, equipping students with tools to analyze and model extreme flood and precipitation risks in urban environments—empowering data-driven decision-making for resilient urban planning.

Fluid Mechanics

Course Assistant

Fall 2025

- Teaching assistant for the undergraduate Fluid Mechanics class. Topics included fluid statics, the Bernoulli equation, control volume analysis, conservation of mass and momentum, the Navier-Stokes equations, dimensional analysis, and viscous flow in pipes and external flows.

- Taught undergraduate students the applications of fluid dynamics to aircraft, spacecrafts, and machinery through lectures and quizzes, covering topics such as Bernoulli’s theorem and the Navier–Stokes equations.

Monitoring Cities

Course Assistant Fall 2025

- Teaching assistant for the graduate Monitoring Cities class. Topics included urban sensing and data acquisition, remote sensing and satellite imagery, aerial and ground-based observation platforms, urban climate and ecology, land use, air quality, and citizen engagement.
- Delivered data science exercises on urban environments using satellite and aerial imagery (Landsat, MODIS, NAIP, Sentinel) to analyze land surface temperature, vegetation, and nighttime lights, as well as LiDAR data to teach digital surface modeling techniques.

Professional Experience

Office of the New York State Comptroller New York, NY, USA
Student Analyst August 2025 to Present

- Assessed the effectiveness of NYC's cooling center program by evaluating indoor thermal comfort and by building a GIS-based coverage analysis that integrated population distribution, the heat vulnerability index, and facility operating hours.

New York University Data Services New York, NY, USA
GIS Specialist May to December 2025

- Delivered ArcGIS lectures and consultations, enhancing patrons’ ability to apply spatial analysis tools effectively across diverse research and institutional projects.

Ministry of the Environment, Japan Tokyo, Japan
Data & Analytics Lead August 2023 to August 2024

- Led a project team deploying robotic automation and genAI to streamline government operations, achieving reduced workload and improved accuracy across 40+ processes.

Deputy Director June 2023 to August 2024

- Organized public meetings between utilities and manufacturers to develop regulatory standards for the digitalization of analog instrumentation in nuclear power plants.

Senior Section Chief May 2017 to June 2023

- Led interministerial consultations to eliminate redundant oversight applied to radiopharmaceuticals under radiation safety and pharmaceutical regulations.

Section Chief April 2015 to May 2017

- Oversaw negotiations with parliament and ministry members for legislative amendments reforming the inspection regime for nuclear plants under IAEA recommendations.

Toshiba Corporation Yokohama, Japan
Research Intern August to September 2013

- Led V&V of a CFD software package and authored technical documentation and user manuals for institutional adoption.

Publications

Yamada, S., & Morooka, S. (2014). Prediction of a boiling transition location and heated surface temperature for upward flow in a vertical tube. Investigation and improvement of prediction

method for fuel rod wall temperature. *Transactions of the Japan Society of Mechanical Engineers*, 80.

(Under Review) Yamada, S., Fitzky, M., & Ghandehari, M. Multiphysics Simulations of Cooling Capacity of Vegetation in the Built Environment. *Urban Climate*. Under Review.

(In Preparation) Yamada, S., & Ghandehari, M. A Spatiotemporal Downscaling of Land Surface Temperature Reshapes Urban Heat Intervention Priorities

Yamada, S., & Ghandehari, M. When Reaching a Cooling Center Helps: A Net-Benefit Accessibility Framework Reveals the Temporal Failure of Urban Cooling Programs.

Conferences Yamada, S., Ghandehari, M., & Fitzky, M. (2025, July). Multi-Physics Analysis of Cooling Effects of Green Spaces: Case Study in Brooklyn, New York. *12th International Conference on Urban Climate*. Rotterdam, the Netherlands.

Yamada, S., & Morooka, S. (2015, March). Study on the prediction of critical power for tubes with obstacles. *Atomic Energy Society of Japan, 2015 Spring Annual Meeting*. Hitachi, Japan.

Yamada, S., & Morooka, S. (2014, December). Study on the prediction of surface temperature for vertical tubes with non-uniform axial heating. *10th International Topical Meeting on Nuclear Thermal Hydraulics, Operation and Safety*. Ginowan, Japan

Yamada, S., & Morooka, S. (2014, November). Development of a droplet transport model for tubes with obstacles. Atomic Energy Society of Japan, Kanto Koetsu Branch, *The 13th Young Researcher Conference*. Tokyo, Japan.

Yamada, S., Yamada, M., & Morooka, S. (2013, June). Study on the prediction of boiling transition and surface temperature for heating vertical tubes. *The Japan Society of Mechanical Engineers, The 18th National Symposium on Power and Energy Systems*. Chiba, Japan.

Yamada, S., & Morooka, S. (2013, March). Prediction of boiling transition and surface temperature for vertical tubes. *Atomic Energy Society of Japan, Kanto Koetsu Branch, The 7th Student Research Conference*. Tokyo, Japan.

Invited Talks

Introduction to GeoAI

Guest Speaker

Fall 2025

- Delivered an invited lecture on the principles, utility, and applications of GeoAI as part of a GIS Day event.
- Conducted a demonstration of machine learning techniques for land-use classification using ArcGIS Pro for a multidisciplinary group of undergraduate and graduate students.

Introduction to Nuclear Science and Engineering

Guest Lecturer

Spring 2019 to 2021

- Delivered lectures that fostered interdisciplinary learning while equipping students with both foundational and advanced knowledge in nuclear system design and regulatory policy.
- Guided students to critically engage with complex technical and policy frameworks, contributing to the development of the next generation of nuclear professionals.

Skills

- **Language:** Fluent in English and Japanese

- **Technical:** Python, C, R, Visual Basic for Applications, STATA, HTML, CSS, JavaScript, ArcGIS Suite, QGIS, Google Earth Engine, Ansys Fluent, OpenFOAM.
- **Professional:** Environmental laws & regulations, human resource management.

**Outreach
Efforts**

- Founder, American Society for Photogrammetry and Remote Sensing, NYU Chapter.